

Vowels detection using Decision Trees (Machine Learning)

Miguel Roca

Universidad CEU San Pablo, Montepríncipe

miguel.rocatorres@usp.ceu.es

Abstract

This study addresses the critical challenge of automatic vowel detection in adverse acoustic environments, where real-world noise severely degrades conventional speech recognition systems. To build acoustic resilience, we investigate the evolution of a Decision Tree classifier trained on Mel-Frequency Cepstral Coefficients (MFCCs). Rather than relying on a fragile laboratory baseline, the methodology focuses on extensive Data Augmentation and dynamic noise injection to simulate search and rescue (SAR) scenarios. Furthermore, this paper analyzes the spectral mismatch induced by traditional denoising filters, proving their inefficiency with dynamic acoustic features. Finally, Out-of-Distribution (OOD) field testing reveals the algorithm's high empirical variance in chaotic conditions, ultimately demonstrating the mathematical limitations of decision boundaries and proposing deep neural architectures for future work.

Index Terms: speech recognition, MFCC, Decision Trees, data augmentation, spectral mismatch, deep learning.

1. Introduction

The rapid identification of survivors in search and rescue (SAR) operations relies heavily on acoustic signals, yet traditional speech recognition often fails under extreme environmental interference. As illustrated in Figure 1, each vowel possesses a distinct spectral envelope across the first 13 Mel-Frequency Cepstral Coefficients (MFCCs) in clean conditions [1]; however, real-world noise severely disrupts these natural geometric boundaries. To address this, this paper investigates the evolution of a Decision Tree classifier [2] for robust vowel detection.

To tackle the complexity of real-world interference, this study expands the traditional baseline by extracting 39 dynamic acoustic features—including first and second-order derivatives (Deltas and Delta-Deltas)—and scaling the dataset to over 14,000 samples through Pitch Shifting, Time Stretching, and dynamic Signal-to-Noise Ratio (SNR) injection. We empirically analyze how these mathematical adjustments interact with the classifier's geometry, contrasting them with the spectral mismatch typically induced by standard denoising filters.

To document this progression, the remainder of this report is structured as follows. Section 2 details the acoustic feature extraction. Section 3 presents the initial methodology and baseline results. Section 4 explains the successful data augmentation strategies. Section 5 analyzes the failed optimization attempts. Finally, Section 6 evaluates the model's high variance during Out-of-Distribution field tests and concludes the study.

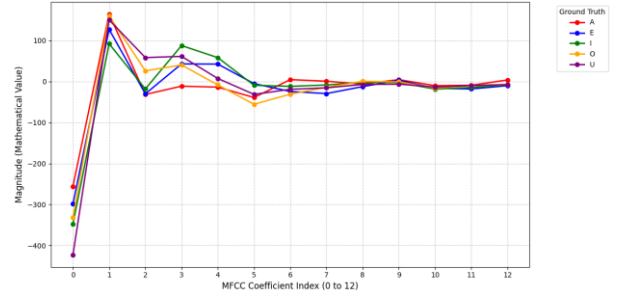


Figure 1: Average MFCC Acoustic Signature per Spanish Vowel

2. Acoustic Feature Extraction

For a Machine Learning model to process audio signals, it is imperative to transform the raw sound wave into a mathematical representation that highlights relevant phonetic characteristics. In this project, Mel-Frequency Cepstral Coefficients (MFCCs) were selected as the primary acoustic feature. These coefficients are the standard in speech recognition because they apply a logarithmic transformation (the Mel scale) that non-linearly simulates the auditory perception of the human ear, prioritizing the low and mid-frequencies where vowel formants reside [1].

2.1. The MFCC Extraction Pipeline

Before computing the dynamic derivatives, the raw audio signal undergoes a rigorous signal processing pipeline to isolate the static MFCCs. First, a pre-emphasis filter is applied to amplify the high frequencies, compensating for the natural human tendency to produce weaker high-frequency sounds. Next, the continuous audio stream is divided into short overlapping frames (typically 20 to 30 milliseconds) using a Hamming window to prevent signal discontinuities at the edges.

A Fast Fourier Transform (FFT) is then executed to convert the signal from the time domain into the frequency domain. The resulting spectrum is wrapped through a bank of triangular Mel-scale filters, which map the frequencies to human auditory perception. Finally, a Discrete Cosine Transform (DCT) is applied to the log-Mel spectrum. The DCT compresses the acoustic information and decorrelates the filterbank energies, yielding the highly independent baseline MFCCs used by the classifier.

2.2. Dynamic Features: Deltas and Delta-Deltas

However, speech is not a static phenomenon but a highly dynamic process. Relying solely on static MFCCs—which represent a "snapshot" of an audio frame—proved insufficient for capturing sound transitions in noisy environments. Therefore, the feature vector was expanded to include the first

and second-order derivatives of the MFCCs, mathematically known as Deltas (Δ) and Delta-Deltas ($\Delta\Delta$) [3].

- Deltas (Δ): Calculate the "velocity" at which the MFCCs change between consecutive frames, capturing the trajectory of the vocal tract.
- Delta-Deltas ($\Delta\Delta$): Calculate the "acceleration" of these changes, providing critical information about the abruptness of vocal attacks and transitions.

By combining 13 base MFCCs with their 13 Deltas and 13 Delta-Deltas, a robust 39-dimensional feature vector was generated for each audio segment, ensuring a comprehensive geometric representation of the vowels before training the classifier.

3. Methodology

The experimental workflow was designed to ensure that the classifier could transition from pristine laboratory conditions to chaotic real-world scenarios. This section details the initial data acquisition, the model architecture, and its preliminary performance.

3.1. Data Acquisition and Recording Protocol

The primary dataset was constructed to establish a high-fidelity baseline. Participants were instructed to produce the five Spanish vowels in a sustained and clear manner. For precise acoustic framing, these target sounds correspond to the standard International Phonetic Alphabet (IPA) transcriptions for Spanish phonetics [4]: /a/ (open central), /e/ (close-mid front), /i/ (close front), /o/ (close-mid back), and /u/ (close back). To minimize ambient interference, all recordings were conducted in a controlled, silent environment. This resulted in a clean corpus of 165 original samples, which served as the "ground truth" for the preliminary training phase.

3.2. Classification Architecture

The system was implemented using Python and the scikit-learn library [5]. A Decision Tree classifier was selected due to its logical transparency and ability to handle non-linear decision boundaries. The algorithm was configured to process the 39-dimensional acoustic feature vectors (MFCCs and Deltas) detailed in Section 2. To establish an initial baseline without complex interventions, basic pre-pruning hyperparameters were applied.

3.3. Baseline Results and Noise Vulnerability

The Decision Tree was first trained and evaluated exclusively on the 165 clean samples. In this noise-free environment, the system achieved a baseline accuracy of 83.00%, successfully recognizing most of the phonetic patterns, as illustrated in Figure 2. However, to assess its acoustic resilience, this laboratory-trained model was subjected to a stress test using a significantly expanded dataset of 14,850 samples, which were synthetically corrupted by environmental noise.

Under these conditions, the accuracy drastically collapsed to 49.33%. As shown in Figure 3, the introduction of noise particularly devastated the recognition of vowels 'I', 'O', and 'U', demonstrating how overlapping frequencies destroy the spectral envelope. This sharp decline empirically proved the model's inability to generalize phonetic structures under interference, justifying the advanced training methodologies discussed in Section 4.

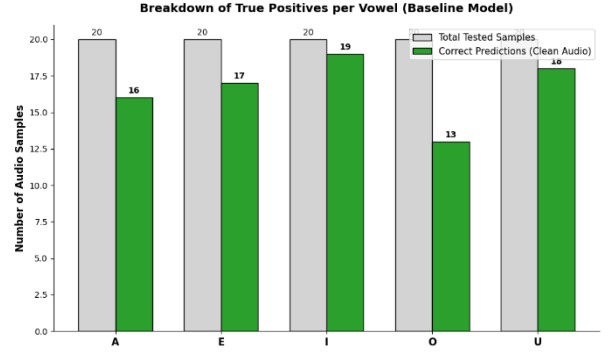


Figure 2: Breakdown of true positives per vowel evaluated under clean laboratory condition.

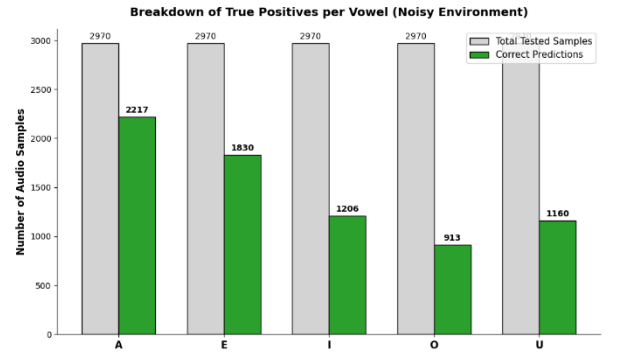


Figure 3: Breakdown of true positives per vowel evaluated under chaotic environmental noise.

4. Robust Training Strategies and Optimization

As demonstrated by the baseline evaluation, the original Decision Tree severely overfitted to the clean acoustic signatures of the laboratory. To bridge the 33.67% performance gap caused by environmental interference, a comprehensive Data Augmentation and dynamic training strategy was implemented. This approach aimed to force the classifier to prioritize human vocal tract resonance over ambient background noise.

4.1. Dynamic Environmental Noise Injection

In real-world Search and Rescue (SAR) operations, acoustic noise is inherently broadband and unpredictable. To simulate this chaotic acoustic environment, the original dataset was digitally mixed with 21 distinct environmental noises retrieved from open-source sound banks, including wind, heavy machinery, sirens, and structural debris. Rather than applying a static overlay, the algorithm dynamically injected these interferences at varying Signal-to-Noise Ratios (SNR). By mathematically altering the amplitude of the background noise relative to the vocal signal, the Decision Tree was forced to identify the underlying MFCC geometric patterns regardless of the noise floor's intensity.

4.2. Data Augmentation Techniques

To further expand the representational capacity of the model and prevent memorization of specific audio clips, structural modifications were applied to the raw waveforms before MFCC extraction [6]. Two primary techniques were utilized:

- **Pitch Shifting:** The fundamental frequency (F0) of the vocalizations was mathematically raised or lowered by fractional semitones. This simulated different vocal tract lengths and physiological variations among potential victims, increasing variance without altering the phonetic identity of the vowel.
- **Time Stretching:** The playback speed of the audio arrays was randomly modified without affecting the pitch. This replicated the varying speaking rates typical of human speech under stress or panic.

Through the combination of dynamic SNR injection, Pitch Shifting, and Time Stretching, the initial dataset of 165 clean recordings was massively scaled to a highly diverse corpus of 14,850 synthetic samples.

4.3. Hyperparameter Tuning and Robust Results

Training a Decision Tree on such a chaotic, high-dimensional dataset requires strict regularization. Without it, the algorithm would recursively split nodes until every single training sample is perfectly isolated, essentially memorizing the ambient noise instead of learning the phonetic rules.

To mitigate this, the node splitting criterion was governed by the Gini Impurity index, which mathematically evaluates the probability of incorrectly classifying a randomly chosen element if it were randomly labeled according to the distribution of labels in the node. To prevent the tree from fitting to stochastic noise patterns, structural hyperparameter limits were enforced. The `max_depth` of the tree was capped to stop deep, noise-specific branches from forming, while `min_samples_split` and `min_samples_leaf` thresholds were introduced to ensure that a split only occurs if it is statistically representative of a broader acoustic trend.

The implementation of these combined strategies yielded a dramatic improvement. When the regularized Decision Tree was trained and evaluated on the augmented dataset of 14,850 noisy samples, the classification accuracy surged to 90.54%. As illustrated in Figure 4, the model achieved high consistency across all vowel classes, successfully isolating the 39-dimensional acoustic features despite the intense environmental interference. This milestone indicated that the system had successfully learned to isolate and map the 39-dimensional acoustic features of the vowels, demonstrating high acoustic resilience within the synthetic training domain.

5. Failed Optimization Attempts and Spectral Mismatch

While the dynamic Data Augmentation strategy proved successful, several traditional audio processing techniques were empirically tested and subsequently discarded during the development phase. Documenting these failed attempts is crucial, as it highlights the geometric sensitivity of MFCCs and the limitations of standard noise-reduction algorithms in Search and Rescue (SAR) contexts.

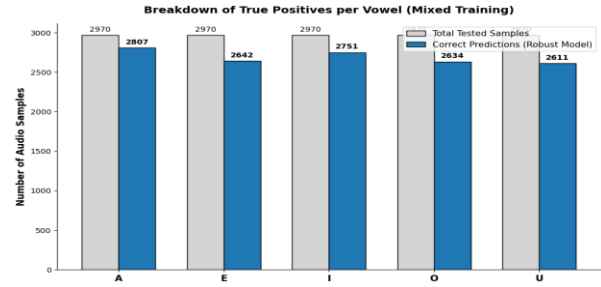


Fig. 4. Breakdown of true positives per vowel evaluated after dynamic data augmentation and mixed-condition training.

5.1. Traditional Band-Pass Filtering

The first intuitive approach to mitigate environmental interference was the application of a digital band-pass filter, designed to isolate the typical human fundamental frequency range (roughly 300 Hz to 3400 Hz). However, real-world SAR noise — such as sirens, wind shear, and heavy machinery — is inherently broadband and overlaps significantly with human phonetic formants. Consequently, the filter not only failed to remove the targeted noise but also attenuated critical high-frequency vocal formants, flattening the signal and confusing the classifier.

5.2. Spectral Gating and Artifact Induction

A secondary attempt utilized the *noisereduce* algorithm, a popular spectral gating technique. This method profiles background noise during "silent" frames and subtracts that spectral footprint from the active speech frames. While this approach is effective for static background hums (like air conditioning), it catastrophically failed with unpredictable dynamic noises. The subtraction process generated non-linear acoustic artifacts, commonly known as "musical noise." These artifacts heavily distorted the MFCC extraction, creating a severe *spectral mismatch* between the clean training data and the filtered testing data.

5.3. Excessive White Noise and Overfitting

Finally, an experiment was conducted where extreme levels of synthetic Gaussian White Noise were injected into the training set in an attempt to make the Decision Tree completely noise-agnostic. Instead of focusing on the vocalizations, the mathematical variance of the white noise overwhelmed the acoustic features. The Decision Tree's nodes adapted to classify the random noise patterns rather than the underlying phonetic structure, leading to severe overfitting and a complete loss of predictive capability on unseen data [7].

Ultimately, these experiments proved that destructive pre-processing (filtering or spectral subtraction) is detrimental when using dynamic MFCCs. The optimal path, as demonstrated in Section 4, was to train the model to "see through" the noise via mixed-condition training rather than attempting to artificially remove it.

6. Out-of-Distribution Validation

To definitively validate the optimized Decision Tree, a final Out-of-Distribution (OOD) field test was conducted. The model, trained exclusively on laboratory and synthetically augmented data, was deployed to classify vowels recorded directly in a real-world highway environment.

6.1. High Variance and the Random Seed Phenomenon

The field test exposed a critical vulnerability of the algorithm: high variance. Despite achieving 90.54% accuracy during the synthetic testing phase, the real-world performance experienced a significant drop. Furthermore, empirical testing revealed that the final accuracy fluctuated drastically depending solely on the random seed initialization used during the model's training phase. Testing distinct seeds resulted in accuracy swinging between a baseline of 40.00% and a peak of 50.00%. This 10% performance gap, dictated entirely by statistical randomness rather than actual acoustic learning, demonstrates that the Decision Tree memorized specific noise splits rather than achieving true phonetic generalization.

This extreme sensitivity to the random seed is fundamentally rooted in the mathematical nature of standard Decision Trees. They utilize a top-down, greedy recursive partitioning algorithm. Because they make the locally optimal split at each node without looking ahead at the global structure, even a minute variation in the initial random distribution of the training data (dictated by the seed) can select a slightly different initial feature for the root node. In chaotic acoustic environments, this entirely alters the subsequent branching topology. Consequently, a tree initialized with one seed might rely heavily on the Delta coefficients of low frequencies, while another might branch based on high-frequency noise artifacts, leading to the observed 10% discrepancy in real-world predictive power.

7. Final Conclusions and Future Work

This paper has presented a comprehensive evaluation of a Decision Tree classifier for automatic vowel detection tailored for Search and Rescue (SAR) emergency scenarios. Recognizing the fragility of laboratory baselines when exposed to hostile acoustic environments, the core methodology investigated the integration of a 39-dimensional feature space—combining baseline MFCCs with first and second-order dynamic derivatives—alongside robust data augmentation techniques including pitch shifting, time stretching, and mixed-condition noise injection.

The empirical results demonstrate a stark contrast between controlled synthetic domains and unobserved real-world deployment. Within the synthetically expanded training domain containing 21 diverse noise types, strict hyperparameter regularization utilizing Gini impurity criteria successfully pushed the model's classification accuracy to a peak of 90.54%. This confirmed that the decision boundaries could effectively map phonetic structures under simulated degradation.

However, the subsequent Out-of-Distribution (OOD) field validation in a chaotic highway environment exposed severe limitations. The model suffered a critical drop in predictive performance, accompanied by an unstable 10% accuracy fluctuation (ranging from 40.00% to 50.00%) tied entirely to random seed initialization. This behavior mathematically exposes the structural vulnerability of the top-down, greedy

recursive partitioning mechanism inherent to Decision Trees, which tends to latch onto stochastic noise artifacts rather than forming robust phonetic representations.

Consequently, while Decision Trees offer low computational latency suitable for edge deployment, their rigid orthogonal decision boundaries lack the generalization capacity required for unpredictable broadband interference. To achieve true acoustic resilience in critical life-saving applications, future work must abandon shallow heuristic classifiers in favor of Deep Neural Network (DNN) architectures [8]. Transitioning to deep convolutional or recurrent models will allow the system to automatically extract non-linear, hierarchical acoustic features from corrupted signals, resolving the high-variance limitation and ensuring reliable performance in real-world disaster zones.

8. Generative AI Disclosure

During the preparation of this work, the author used Google Gemini to assist with language translation, structural formatting, and the refinement of academic prose. After using this tool, the author thoroughly reviewed, edited, and validated all generated content, assuming full responsibility for the final integrity, technical accuracy, and originality of this publication.

9. References

- [1] S. Davis and P. Mermelstein, "Comparison of parametric representations for monosyllabic word recognition in continuously spoken sentences," *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 28, no. 4, pp. 357–366, 1980.
- [2] L. Breiman, J. H. Friedman, R. A. Olshen, and C. J. Stone, *Classification and Regression Trees*, Belmont, CA, USA: Wadsworth, 1984.
- [3] S. Furui, "Speaker-independent isolated word recognition using dynamic features of speech spectrum," *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 34, no. 1, pp. 52–59, 1986.
- [4] J. I. Hualde, *The Sounds of Spanish*. Cambridge, U.K.: Cambridge University Press, 2005.
- [5] B. McFee et al., "librosa: Audio and music signal analysis in python," in *Proc. 14th Python in Science Conference*, Austin, TX, USA, 2015, pp. 18–24.
- [6] J. Salamon and J. P. Bello, "Deep convolutional neural networks and data augmentation for environmental sound classification," *IEEE Signal Processing Letters*, vol. 24, no. 3, pp. 279–283, 2017.
- [7] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, New York, NY, USA: Springer, 2009.
- [8] G. Hinton et al., "Deep neural networks for acoustic modeling in speech recognition," *IEEE Signal Processing Magazine*, vol. 29, no. 6, pp. 82–97, 2012.